STUDYNEST | CLASS 11 PHYSICS

MARKING SCHEME AND MODEL ANSWERS

Chapter 5: Work, Energy and Power

Maximum Marks: 40 | Suggested time: 60 minutes

Use after attempting the paper. Accept equivalent correct methods.

1 mark: Correct fact, formula, unit or direction.
2 marks: Principle/formula (1); explanation, working or result (1).
3 marks: Relation (1); working/reasoning (1); final result and unit (1).
4 marks: Principle (1); setup/equation (1); working (1); result/interpretation (1).
Mark splits are a guide; award partial credit for valid steps.

When force is perpendicular to displacement.

mv^2.

Watt.

Net work equals change in kinetic energy.

kx^2.

Momentum and kinetic energy.

mgh.

Zero.

40 J.

24 J.

2000 J.

0 J.

1 J.

1 m/s.

32 J.

sqrt60approx 7.75 m/s.

18 J.

-200 kJ; 4000 N.

10 m/s; sqrt80approx 8.94 m/s.

2 m/s; 24 J lost.